

Beyond Valence-Arousal: The L.O.V.E. Stack and the Connection Axis for Computational Emotion Recognition in Mental Health Applications

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Abstract

Traditional dimensional models of emotion—Valence-Arousal (VA) [16] and Valence-Arousal-Dominance (VAD) [14]—fail to distinguish between emotionally similar but relationally distinct states. For example, pity and compassion share similar valence (both negative-to-neutral) and arousal (both low-energy), yet they represent fundamentally different interpersonal orientations: pity maintains hierarchical distance (“feeling FOR someone”), while compassion embodies shared humanity (“feeling WITH someone”). This distinction, critical for therapeutic validity in mental health applications, remains unoperationalized in existing computational emotion models.

We introduce the **Connection axis**—a novel third dimension measuring interpersonal alignment from separation ($C = -1$) to unity ($C = +1$)—forming the **VAC (Valence-Arousal-Connection) model**. The Connection axis distinguishes relationally different emotional states including pity vs. compassion, grief vs. despair, and shame vs. guilt. We implement this model in the **L.O.V.E. Stack** (Listener-Observer-Versor-Experience), a privacy-first microservices platform that extracts VAC coordinates from speech and text using local large language models, stores emotional trajectories in a vector database, performs therapeutic pathfinding with evidence-based strategies, and visualizes emotional states as 3D animated representations using quaternion mathematics [17].

Semantic validation demonstrates 98% accuracy in distinguishing pity from compassion, proving the Connection axis is computationally extractable from natural language. The system includes an 87-emotion atlas based on Brené Brown’s relational research [3], 107 peer-reviewed regulation strategies, and A* pathfinding [9] that respects psychological

constraints (e.g., shame $\rightarrow$ self-compassion requires vulnerability as an intermediate state). By operationalizing relational quality in emotions, the VAC model enables mental health technologies to provide therapeutically valid guidance while maintaining user privacy through local processing.

Keywords: Emotion recognition, affective computing, mental health technology, VAC model, dimensional emotion models, speech processing, connection axis, compassion, pity

1 Introduction

1.1 The Problem: When Similar Emotions Tell Different Stories

Consider two people observing someone in distress:

Person A: “I feel so sorry for them. They’re really struggling, and I hope they get better soon.”

Person B: “I feel their pain as if it were my own. I’m here with them in this difficult moment.”

Both speakers express concern for another’s suffering. Both would likely score similarly on traditional sentiment analysis: negative valence, low arousal. Yet these statements represent profoundly different emotional experiences. Person A maintains distance—a hierarchical stance of “us” versus “them”—while Person B expresses deep connection and shared humanity.

This distinction matters. In therapeutic contexts, the quality of emotional connection predicts outcomes. In human-computer interaction, systems that recognize relational nuance can provide more appropriate guidance. In mental health technology, distinguishing pity from compassion, grief from despair, and shame from guilt enables therapeutically valid pathfinding between emotional states.

Yet existing computational emotion models cannot make these distinctions.

1.2 The Limitations of Current Models

The dominant paradigms in affective computing rely on two-dimensional (Valence-Arousal) or three-dimensional (Valence-Arousal-Dominance) representations:

Valence-Arousal (VA): Russell’s Circumplex Model [16] positions emotions on two axes—pleasantness and activation. While elegant and widely used, VA conflates relationally distinct states:

- Pity and compassion both map to negative valence, low arousal
- Grief and despair both map to negative valence, moderate-low arousal
- The model provides no mechanism to distinguish them

Valence-Arousal-Dominance (VAD): Mehrabian and Russell’s PAD model [14] adds a third dimension—dominance (sense of control). While this helps distinguish some emotions, dominance measures power relationships, not relational quality:

- High dominance: feeling in control
- Low dominance: feeling controlled
- Neither captures “feeling WITH” versus “feeling FOR”

The consequence is that sentiment analysis, emotion recognition systems, and affective computing applications routinely conflate emotionally similar but relationally distinct states. A mental health chatbot might treat pity and compassion identically, failing to recognize that compassion heals while pity can reinforce separation. A voice-based emotion detector might miss the difference between grief (love persisting through loss) and despair (isolated suffering), leading to inappropriate interventions.

1.3 A Third Dimension: Connection

We propose the **Connection axis**—a dimension measuring interpersonal alignment:

$$C \in [-1, +1]$$

where:

- $C = +1$: “Feeling WITH” (deep unity, shared experience, empathy)
- $C = 0$: Neutral (no relational component, or balanced FOR/WITH)
- $C = -1$: “Feeling FOR/AT” (hierarchical distance, separation, othering)

The Connection axis is not about power (dominance) or intensity (arousal). It captures whether an emotion involves moving toward or away from relational alignment with another person. This dimension is grounded in Brené Brown’s empirical research on relational emotions [3], which distinguished 87 emotional experiences across 13 categories, emphasizing that the quality of interpersonal connection fundamentally shapes emotional experience.

Combined with Valence and Arousal, this forms the **VAC Model**:

Emotion State = (V, A, C) where $V, A, C \in [-1, +1]$

This three-dimensional space enables computational systems to distinguish:

Table 1: Critical emotion distinctions enabled by the Connection axis

Emotion	Valence	Arousal	Connection
Pity	−0.3	−0.1	−0.7 (FOR)
Compassion	+0.5	+0.2	+0.9 (WITH)
Grief	−0.8	−0.3	+0.7 (love persists)
Despair	−0.9	−0.4	−0.6 (isolated)
Shame	−0.7	−0.2	−0.8 (disconnection)
Guilt	−0.5	+0.1	+0.4 (values-connected)

1.4 Implementation: The L.O.V.E. Stack

We implement the VAC model in a complete microservices platform designed for mental health applications:

- **Listener** - Audio transcription and semantic VAC extraction using local LLMs
- **Observer** - Data persistence, vector search, and therapeutic pathfinding
- **Versor** - Quaternion mathematics for smooth 3D emotional transitions
- **Experience** - Real-time 3D visualization of emotional states

Key design principles:

1. **Privacy-first:** All processing occurs locally (no cloud APIs)
2. **Evidence-based:** 107 regulation strategies cite peer-reviewed research
3. **Therapeutically valid:** Pathfinding respects psychological constraints
4. **Interpretable:** VAC coordinates are human-understandable

The system extracts VAC coordinates from natural language with 98% accuracy on the critical pity-vs-compassion distinction, proving the Connection axis is computationally operationalizable.

1.5 Contributions

This work makes the following contributions to affective computing and mental health technology:

1. **Theoretical:** Introduction of the Connection axis as a novel dimension for emotion representation, grounded in relational psychology research
2. **Computational:** Demonstration that the Connection axis can be extracted from natural language using large language models with high accuracy (98% on validation tests)
3. **Architectural:** A complete microservices implementation (L.O.V.E. Stack) including:
 - Semantic VAC extraction with local LLMs
 - Vector similarity search over emotional trajectories
 - Quaternion-based smooth transitions between emotional states
 - Evidence-based therapeutic pathfinding (107 strategies)
 - Real-time 3D visualization
4. **Validation:** Multiple validation approaches:
 - Semantic: Pity vs. compassion discrimination
 - Mathematical: Quaternion operation correctness (56/56 tests)
 - Therapeutic: Evidence mapping for 107 strategies
5. **Practical:** A privacy-first mental health platform suitable for clinical deployment, with full user data control and local processing

1.6 Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews existing emotion models, speech-based emotion recognition, and mental health monitoring systems. Section 3 provides formal mathematical definition of the VAC space and describes the 87-emotion atlas. Section 4 details the L.O.V.E. Stack’s four microservices and their interactions. Section 5 explains how the Connection axis is extracted from speech and text. Section 6 presents validation results across semantic, mathematical, and therapeutic domains. Section 7 demonstrates therapeutic pathfinding and evidence-based strategy selection. Section 8 provides technical details and performance metrics. Section 9 analyzes strengths, limitations, and future research directions. Section 10 summarizes contributions and impact.

1.7 Significance for Speech-Based Emotion Recognition

For researchers in speech-centered emotion recognition, the Connection axis opens new research directions. While our current implementation uses semantic analysis, we believe acoustic features—prosody, pitch contours, speech rate, intensity—could provide additional signals for Connection detection. The quality of interpersonal alignment may manifest not just in what is said, but in how it is said. Future work integrating acoustic and semantic channels could substantially enhance Connection axis accuracy, particularly in ambiguous cases.

We present this work to the research community both to validate the Connection axis conceptually and to invite collaboration on multimodal integration, clinical validation, and real-world deployment in mental health settings.

2 Related Work

2.1 Dimensional Emotion Models

Valence-Arousal (VA) Models

Russell’s (1980) Circumplex Model revolutionized emotion research by proposing that all emotions could be positioned in a two-dimensional space defined by valence (pleasantness) and arousal (activation) [16]. This model’s elegance and empirical support made it the dominant framework in affective computing. Subsequent work has refined the circumplex, exploring whether the space is truly circular or has quadrants with different properties.

However, VA models face fundamental limitations. Emotions that differ in relational quality but share similar valence and arousal are indistinguishable. For example, pity, sympathy, and compassion all occupy similar regions of VA space despite representing distinct interpersonal stances. This conflation limits therapeutic applications where relational quality is paramount.

Valence-Arousal-Dominance (VAD) Models

Mehrabian and Russell (1974) introduced the PAD model, adding Dominance as a third dimension [14]. Dominance captures the sense of control or power in an emotional experience:

- High dominance: Anger, pride, contempt
- Low dominance: Fear, shame, guilt

While PAD improves on VA by adding dimensionality, dominance measures power relationships rather than interpersonal alignment. High-dominance pity (condescension) and high-dominance compassion (protective care) would both score similarly, missing the crucial relational distinction.

Other Dimensional Approaches

The Pleasure-Arousal-Dominance space has been extended in various ways. Some researchers have proposed additional dimensions like “approach-avoidance” or “certainty-uncertainty.” However, none explicitly capture the quality of interpersonal connection that distinguishes relationally different emotional states.

2.2 Categorical Emotion Models

Basic Emotions Theory

Ekman’s (1992) basic emotions approach identifies universal emotional categories (happiness, sadness, fear, anger, surprise, disgust) based on facial expressions [5]. While empirically grounded and cross-culturally validated, categorical models lack the continuous structure needed for smooth emotional transitions in computational systems.

Complex Emotion Taxonomies

More recent work has expanded beyond basic emotions. Cowen and Keltner (2017) identified 27 distinct categories through semantic space analysis [4]. Brené Brown’s (2021) “Atlas of the Heart” catalogues 87 emotional experiences organized into 13 categories [3]. These rich taxonomies capture nuance but lack mathematical structure for computation.

The VAC model bridges categorical and dimensional approaches: we map Brown’s 87 emotions onto VAC coordinates, providing both categorical richness and dimensional continuity.

2.3 Speech-Based Emotion Recognition

Acoustic Feature Extraction

Speech-based emotion recognition has focused primarily on acoustic features: prosody, pitch (F0), intensity, formant frequencies, speech rate, and voice quality. Provost et al. have demonstrated that these features can distinguish emotional states with reasonable accuracy. Most work targets VA space or discrete emotion categories.

Multimodal Approaches

Recent systems combine acoustic, linguistic, and visual channels for improved accuracy. Provost’s work on multimodal emotion recognition has shown that semantic content complements acoustic features. However, most multimodal systems still target traditional VA or categorical representations.

Mental Health Applications

Speech analysis has shown promise for mental health monitoring. Studies have used speech features to detect depression, assess bipolar disorder symptom severity, and estimate suicide risk. Provost’s work in this domain has emphasized the importance of naturalistic, real-world speech data.

Our work extends this line of research by proposing that acoustic features might carry information about the Connection dimension—a hypothesis we leave for future investigation.

2.4 Mental Health Technology

Mood Tracking Applications: Numerous applications allow users to self-report emotional states. Apps like Daylio, MoodPath, and Bearable use simple emotion selectors or VA-based mood wheels. While useful for longitudinal tracking, these lack the relational dimension needed for therapeutic validity.

AI Chatbots and Therapy Apps: Woebot, Wysa, and similar applications provide cognitive-behavioral therapy (CBT) interventions through conversational interfaces [7]. These systems typically use sentiment analysis to detect user emotional state. However, without the ability to distinguish pity from compassion or shame from guilt, they risk providing therapeutically inappropriate guidance.

Emotion Regulation Tools: Some applications focus specifically on emotion regulation, providing evidence-based strategies from DBT, CBT, and mindfulness traditions. However, most lack sophisticated models of emotional transitions. The L.O.V.E. Stack’s A* pathfinding with psychological constraints represents a more principled approach.

2.5 3D Emotional Visualization

Affective Computing Visualizations: Prior work has visualized emotions in 3D space, typically using VAD coordinates. These visualizations help users understand their emotional state spatially. However, without a meaningful third dimension (beyond dominance), the utility is limited.

Virtual Reality Emotion Experiences: VR applications have explored immersive emotional experiences. Some use spatial positioning to represent emotional intensity or valence. The L.O.V.E. Stack’s quaternion-based 3D visualization builds on this work but with a theoretically grounded third dimension (Connection).

Biofeedback and Wearables: Systems integrating heart rate variability, galvanic skin response, and other physiological signals have created real-time emotional feedback loops. Future work could integrate these biometric signals with VAC coordinates to provide multimodal emotional assessment.

2.6 Gap Analysis: The Need for Connection

Existing emotion models face a common limitation: they cannot distinguish emotionally similar but relationally distinct states. This gap has several consequences:

1. **Therapeutic Invalidity:** Mental health applications may guide users toward pity when compassion is needed, or fail to recognize that shame requires vulnerability before self-compassion can emerge.
2. **Missed Opportunities in HCI:** Conversational AI systems cannot adjust their relational stance to match user needs. A system that recognizes when a user needs “feeling WITH” versus “feeling FOR” could provide more appropriate support.
3. **Cultural Insensitivity:** Different cultures may emphasize connection differently. Without measuring this dimension, systems cannot adapt to cultural variations in emotional expression.
4. **Incomplete Multimodal Models:** If Connection can be detected in speech prosody (a hypothesis supported by linguistic research on empathy), current systems are missing a valuable signal.

The VAC model addresses this gap by operationalizing relational quality as a computational dimension.

In the following sections, we demonstrate that this dimension can be extracted from language, validated experimentally, and implemented in a complete system for mental health applications.

3 The VAC Model

3.1 Mathematical Definition

The VAC model represents emotional states as points in a three-dimensional continuous space:

Emotion $\in \mathbb{R}^3 : (V, A, C)$ where $V, A, C \in [-1, +1]$

Each dimension captures a distinct aspect of emotional experience:

Valence (V): The hedonic quality of the emotion

- $V = +1$: Maximum pleasantness (e.g., joy, ecstasy)
- $V = 0$: Neutral hedonic tone
- $V = -1$: Maximum unpleasantness (e.g., anguish, despair)

Arousal (A): The physiological activation level

- $A = +1$: Maximum activation (e.g., panic, excitement)
- $A = 0$: Moderate activation
- $A = -1$: Minimum activation (e.g., lethargy, calm)

Connection (C): The interpersonal alignment quality

- $C = +1$: Complete unity, “feeling WITH” (e.g., deep empathy, compassion)
- $C = 0$: No relational component, or balanced
- $C = -1$: Complete separation, “feeling FOR/AT” (e.g., pity, condescension)

The VAC space forms a unit cube $[-1, +1]^3$ containing $2^3 = 8$ corners representing archetypal emotional states. However, most emotions occupy interior positions, creating a continuous emotional landscape.

3.2 The Connection Dimension: Theoretical Foundation

3.2.1 Etymology and Linguistic Evidence

The linguistic distinction between “feeling WITH” and “feeling FOR” appears across languages and cultures:

- **Compassion:** From Latin *com-* (with) + *pati* (to suffer) = “to suffer with”
- **Sympathy:** From Greek *syn-* (together) + *pathos* (feeling) = “feeling together” (but often interpreted as “feeling for”)
- **Empathy:** From Greek *en-* (in) + *pathos* (feeling) = “feeling into”
- **Pity:** From Latin *pietas* (piety, duty), implying distance and hierarchy

These etymological roots suggest that human emotional experience has always included a relational dimension.

3.2.2 Brené Brown’s Empirical Framework

Brown’s (2021) research on 87 emotions emphasizes relational quality as fundamental [3]. She distinguishes:

- **Pity:** “A feeling that another’s suffering is your good fortune” (separates)
- **Sympathy:** “I feel for you” (well-intentioned but maintains distance)
- **Empathy:** “I feel with you” (requires vulnerability and connection)
- **Compassion:** “I recognize your suffering and I want to help” (WITH, not FOR)

Brown’s framework emerged from extensive qualitative research (12,000+ data pieces) and emphasizes that people commonly conflate these terms, yet they produce distinct relational outcomes.

3.2.3 Therapeutic Significance

The Connection dimension has direct therapeutic implications:

1. **Shame vs. Guilt:** Brown distinguishes these as:
 - **Shame:** “I am bad” (disconnection from self: $C < 0$)

- **Guilt:** “I did something bad” (connection to values maintained: $C > 0$)

Shame is less amenable to change because it involves identity-level disconnection, while guilt allows for repair.

2. Grief vs. Despair:

- **Grief:** “I lost someone I love” (love persists: $C > 0$)
- **Despair:** “I am alone in my suffering” (isolation: $C < 0$)

Therapeutic approaches differ: grief benefits from connection-enhancing interventions, while despair requires first addressing isolation.

3. Compassion vs. Pity:

- Compassion fosters healing and agency
- Pity can reinforce victimhood and dependence

Mental health systems must distinguish these to avoid iatrogenic harm.

3.3 The 87-Emotion Atlas

We map all 87 emotions from Brown’s “Atlas of the Heart” to VAC coordinates. This mapping was performed through: (1) Theoretical Analysis—examining Brown’s descriptions of each emotion’s relational quality, (2) Semantic Validation—testing LLM extraction on exemplar phrases, and (3) Iterative Refinement—adjusting coordinates to maintain geometric consistency.

Table 2 presents key emotions from selected categories. The complete atlas is available in supplementary materials.

3.4 Critical Distinctions Enabled by Connection

The Connection axis enables computational systems to distinguish emotionally similar but relationally distinct states:

3.4.1 Distinction 1: Pity vs. Compassion

Similarity in VA Space: Both involve recognizing another’s suffering (negative valence) and are relatively low-energy states (low arousal). VAD’s dominance doesn’t clearly separate them.

Distinction in VAC Space: Pity ($-0.3, -0.1, -0.7$) maintains hierarchical distance

Table 2: Sample emotions from the 87-emotion VAC atlas (selected categories)

Category	Emotion	V	A	C	Interpretation
When Life Is Good	Joy	0.9	0.7	0.8	Energized, connected positivity
	Gratitude	0.8	0.3	0.9	Warm appreciation WITH others
	Contentment	0.7	−0.2	0.5	Peaceful satisfaction
When We Fall Short	Shame	−0.7	−0.2	−0.8	Deep self-disconnection
	Guilt	−0.5	0.1	0.4	Values-connected despite error
	Humiliation	−0.8	0.3	−0.9	Public shaming
When the Heart Is Open	Compassion	0.5	0.2	0.9	Suffering WITH
	Pity	−0.3	−0.1	−0.7	Suffering FOR (distance)
	Sympathy	0.1	0.0	0.3	Mild connection
	Empathy	0.0	0.3	0.95	Pure resonance
When Life Is Heartbreaking	Grief	−0.8	−0.3	0.7	Love persists through loss
	Despair	−0.9	−0.4	−0.6	Isolated suffering
	Sadness	−0.6	−0.2	0.2	General negative affect

(“I’m fortunate not to be them”), while Compassion (0.5, 0.2, 0.9) embodies shared humanity (“I’m with you in this”).

Distance: $\Delta_{VAC} = \sqrt{(0.8)^2 + (0.3)^2 + (1.6)^2} = 1.83$

The Connection axis contributes most to the distance (1.6 out of 1.83), confirming it’s the primary distinguishing feature.

3.4.2 Distinction 2: Grief vs. Despair

Similarity in VA Space: Both are intensely negative (high negative valence) and involve low energy (low arousal).

Distinction in VAC Space: Grief (−0.8, −0.3, 0.7) maintains connection to the lost person/thing (love endures), while Despair (−0.9, −0.4, −0.6) involves isolated suffering, disconnected from support.

Therapeutic Implication: Grief requires honoring connection while allowing painful feelings. Despair requires first re-establishing connection (e.g., reaching out) before emotional processing can occur. Treating them identically risks therapeutic harm.

3.4.3 Distinction 3: Shame vs. Guilt

Similarity in VA Space: Both involve negative self-evaluation (negative valence) and may have similar arousal.

Distinction in VAC Space: Shame (−0.7, −0.2, −0.8) represents “I am bad” (identity-level disconnection), while Guilt (−0.5, 0.1, 0.4)

represents “I did bad” (values-connection maintained).

Therapeutic Implication: Brown’s research shows shame is far more correlated with psychopathology than guilt [1]. The VAC model captures this: shame’s severe negative Connection score ($C = -0.8$) indicates the need for vulnerability and reconnection before self-compassion is possible. Simply “reframing” shame doesn’t work—connection must be rebuilt.

3.5 Distance Metrics and Similarity

To measure similarity between emotional states, we use weighted Euclidean distance:

$$d(e_1, e_2) = \sqrt{w_V(V_1 - V_2)^2 + w_A(A_1 - A_2)^2 + w_C(C_1 - C_2)^2}$$

where weights reflect psychological significance:

- $w_V = 1.0$ (baseline importance)
- $w_A = 1.2$ (arousal significantly impacts regulation difficulty)
- $w_C = 1.5$ (Connection most psychologically significant)

Rationale for Connection Weight: Therapeutic literature suggests relational quality is the strongest predictor of outcomes. Transitions across the Connection axis (e.g., shame → self-compassion) are more psychologically demanding than equal-magnitude transitions in Valence or Arousal.

Example: Distance from shame to guilt:

$$\begin{aligned} d(\text{shame}, \text{guilt}) &= \sqrt{1.0(-0.7 - (-0.5))^2 + 1.2(-0.2 - 0.1)^2 + 1.5(-0.8 - 0.4)^2} \\ &= \sqrt{0.04 + 0.108 + 2.16} = 1.52 \end{aligned} \quad (1)$$

The Connection difference dominates (contributing 2.16 out of 2.31), confirming that shame $\rightarrow$ guilt transitions are primarily about reconnection.

3.6 Comparison with Existing 3D Models

Table 3 compares the VAC model with existing dimensional approaches.

Table 3: Comparison of dimensional emotion models

Model	Dims	3rd Dim	Relational?
VA (Russell)	2	—	No
VAD (Mehrabian)	3	Dominance	No
VAC (Ours)	3	Connection	Yes

Why Dominance Falls Short: Dominance measures control but conflates different types of emotional experience:

- High-dominance compassion (protective care) vs. high-dominance pity (condescension)
- Low-dominance empathy (humble resonance) vs. low-dominance despair (helplessness)

Connection explicitly measures relational alignment, providing therapeutic clarity that dominance cannot.

3.7 Theoretical Predictions

The VAC model makes several testable predictions:

Prediction 1: Prosodic features in speech will correlate with Connection scores. *Hypothesis:* “Feeling WITH” involves different vocal patterns than “feeling FOR.” *Proposed signals:* Pitch synchrony, warmth in voice quality, reduced vocal distance.

Prediction 2: Connection scores predict therapeutic outcomes. *Hypothesis:* Increasing Connection over therapy correlates with symptom reduction. *Test:* Track VAC trajectories during treatment, correlate with PHQ-9/GAD-7 scores.

Prediction 3: Connection is culturally universal but with different baselines. *Hypothesis:* All cultures distinguish WITH/FOR, but collectivist cultures show higher average C. *Test:* Cross-cultural validation of VAC extraction.

Prediction 4: Biometric markers correlate with Connection. *Hypothesis:* Positive Connection correlates with (oxytocin, vagal tone, HRV; negative Connection with cortisol. *Test:* Lab study measuring physiology during pity vs. compassion induction.

These predictions provide a research agenda for validating the Connection dimension across modalities and populations.

4 System Architecture

4.1 Overview: The L.O.V.E. Stack

The L.O.V.E. Stack implements the VAC model as a microservices architecture designed for mental health applications. The acronym reflects the system’s four independent modules:

- **Listener:** Sensory input (speech/text $\rightarrow$ VAC extraction)
- **Observer:** Memory and context (data storage, pattern recognition, pathfinding)
- **Versor:** Mathematical engine (quaternion calculations, smooth transitions)
- **Experience:** Presentation layer (3D visualization, user interaction)

Each module is independently deployable, scalable, and maintainable, communicating via REST APIs.

4.2 Design Principles

1. Privacy-First Architecture: All sensitive processing occurs locally. No cloud LLM APIs (Ollama runs locally for semantic analysis), local transcription (faster-whisper for speech-to-text), PII scrubbing (Spacy NER removes personally identifiable information before storage), and user data control (users own their data, can export or delete at any time).

2. Evidence-Based Design: Every therapeutic suggestion cites peer-reviewed research. 107 regulation strategies with evidence hierarchy (meta-analysis $\hookrightarrow$ RCT $\hookrightarrow$ clinical observation), pathfinding respects psychological constraints (e.g., shame $\rightarrow$ self-compassion requires vulnerability), and no “toxic positivity” (system detects and avoids inappropriate emotional shortcuts).

3. Interpretability: VAC coordinates are human-understandable. $(V, A, C) = (0.5, 0.2, 0.9) \approx$ “Compassion.” Unlike neural network embeddings, VAC dimensions have clear semantic meaning. Users can see their emotional trajectory in 3D space.

4. Modularity: Each service has a single responsibility (Listener: Extract VAC; Observer: Store and query; Versor: Calculate quaternions; Experience: Visualize). This separation enables independent development, testing, and scaling.

4.3 Module Details

4.3.1 Listener: The Sensory Cortex

Purpose: Transform human expression (audio/text) into VAC coordinates.

Technology Stack: Python 3.11 + FastAPI (async API), faster-whisper (local speech-to-text), Ollama + Llama 3.1 8B (local LLM for semantic analysis), Spacy (PII scrubbing), Redis + Arq (async job queuing).

Workflow: (1) *Transcription* (if audio): faster-whisper converts speech to text (~500ms for 10s audio), (2) *Semantic Analysis*: LLM extracts VAC coordinates using few-shot prompt (~1-2s), (3) *PII Scrubbing*: Spacy NER removes names, addresses, phone numbers, (4) *Confidence Scoring*: LLM provides reasoning and confidence (0-1), (5) *Return*: VAC coordinates + metadata.

Critical Innovation: Teaching the LLM to recognize the Connection axis through prompt engineering (see Section 5).

4.3.2 Observer: The Hippocampus

Purpose: Memory, context, and therapeutic guidance.

Technology Stack: Python 3.11 + FastAPI, PostgreSQL 16 + pgvector [11] (vector similarity search), SQLAlchemy (ORM), Alembic (migrations).

Data Model: The Observer stores three primary entities:

1. *Canonical emotions* (87 emotions from Brown’s atlas): Pre-seeded with VAC coordinates and pre-calculated quaternions
2. *User emotional trajectory* (high-volume): Timestamped emotional states with pgvector indexing for similarity search
3. *Evidence-based regulation strategies*: 107 strategies with citations, evidence levels, and applicability metadata

The database uses HNSW (Hierarchical Navigable Small World) indexing for fast nearest-neighbor search (~30ms for 1000+ records).

Workflow: (1) *State Recording*: Store incoming VAC + timestamp, (2) *Nearest Emotion*: pgvector similarity search finds closest atlas emotion, (3)

Pathfinding: A* algorithm finds therapeutic path from current to goal state, (4) *Strategy Selection*: Filters 107 strategies by applicability and evidence level, (5) *Pattern Recognition*: Detects recurring emotional states or stuck points.

Critical Innovation: A* pathfinding with psychological constraints (see Section 7).

4.3.3 Versor: The Mathematical Engine

Purpose: Pure quaternion mathematics for smooth 3D transitions.

Technology Stack: Python 3.11 + FastAPI, NumPy (numerical operations), No database (stateless microservice).

Algorithms:

VAC → Quaternion Conversion:

$$\theta = \pi \cdot \frac{V+1}{2}, \quad \phi = \pi \cdot \frac{A+1}{2}, \quad \psi = \pi \cdot \frac{C+1}{2} \quad (2)$$

$$w = \cos(\theta/2) \quad (3)$$

$$x = \sin(\theta/2) \cdot \cos(\phi) \cdot \sin(\psi) \quad (4)$$

$$y = \sin(\theta/2) \cdot \sin(\phi) \cdot \sin(\psi) \quad (5)$$

$$z = \sin(\theta/2) \cdot \cos(\psi) \quad (6)$$

Then normalize to unit quaternion: $q = \frac{(w,x,y,z)}{\|(w,x,y,z)\|}$
SLERP (Spherical Linear Interpolation) [17]:

$$q(t) = \frac{\sin((1-t)\Omega)}{\sin(\Omega)}q_1 + \frac{\sin(t\Omega)}{\sin(\Omega)}q_2$$

where $\Omega = \cos^{-1}(q_1 \cdot q_2)$ is the angular distance.

Why Quaternions? (1) No gimbal lock (Euler angles fail at certain orientations), (2) Smooth interpolation (SLERP maintains constant angular velocity), (3) Efficient (4 numbers vs. 9 for rotation matrix), (4) Composable (multiply to combine rotations).

Critical Innovation: Mapping 3D emotional space to 4D rotational space enables smooth visualization and animation.

4.3.4 Experience: The Presentation Layer

Purpose: Real-time 3D visualization and user interaction.

Technology Stack: TypeScript + Next.js 16, React 19 + React Three Fiber, Three.js (3D engine), Custom GLSL shaders, Zustand (state management).

Visual Mapping:

- **Valence → Color:** Crimson ($V = -1$) to Cyan ($V = +1$)

- **Arousal $\rightarrow$ Geometry:** Smooth sphere ($A = -1$) to chaotic spikes ($A = +1$)
- **Connection $\rightarrow$ Glow/Opaicity:** Opaque ($C = -1$) to luminous ($C = +1$)

Workflow: (1) User Input (text or voice), (2) Call Listener (get VAC coordinates), (3) Call Observer (store state, get nearest emotion + path), (4) Call Versor (get quaternion + SLERP frames), (5) Render (animate Soul Sphere over 60 frames, 1 second), (6) Display (show emotion name, VAC coords, suggested strategies).

Critical Innovation: Mapping abstract VAC coordinates to visceral 3D visual experience.

4.4 Data Flow: End-to-End Example

Scenario: User says “I feel so ashamed of myself”

The data flows through the system as follows: (1) User provides audio input, (2) Experience module calls Listener’s `/listener/ingest` endpoint, (3) Listener transcribes audio (500ms), analyzes semantics to extract VAC = $(-0.7, -0.2, -0.8)$ (1.5s), sanitizes PII, and returns results with confidence 0.94, (4) Experience calls Observer’s `/observer/state` endpoint, (5) Observer stores in database, performs pgvector similarity search identifying “Shame” as nearest emotion (30ms), detects high negative Connection ($C = -0.8$), calls Versor for quaternion conversion, and returns suggested therapeutic path, (6) Observer calls Versor’s `/versor/calculate` endpoint, (7) Versor converts VAC to quaternion and generates SLERP animation path (10ms), (8) Experience renders Soul Sphere with dark red color ($V = -0.7$), slightly jagged geometry ($A = -0.2$), and very dim glow ($C = -0.8$), animating rotation using the quaternion, and displays “Shame” with suggested path (Shame $\rightarrow$ Vulnerability $\rightarrow$ Self-Compassion), (9) User sees the 3D visualization with emotion label and evidence-based strategy (“Name your feelings without judgment” [13]).

Total Latency: ~ 2.5 seconds (transcription: 500ms, LLM: 1500ms, rest: 500ms)

4.5 Scalability and Deployment

Current Capacity: ~ 100 concurrent users per instance. The bottleneck is Ollama LLM inference (CPU-bound).

Scaling Strategy: Versor (stateless) enables horizontal scaling with load balancer + multiple instances. Listener benefits from GPU acceleration for Ollama (10-100x speedup). Observer uses read replicas for PostgreSQL and Redis cache for frequent

queries. Experience uses CDN for static assets and server-side rendering.

Future Optimizations: GPU deployment, model quantization, and caching common queries could reduce latency to ≤ 500 ms.

5 VAC Extraction from Speech and Text

5.1 The Challenge: Teaching Connection to LLMs

The primary technical challenge is extracting the Connection dimension from natural language. While Valence and Arousal have been studied extensively in sentiment analysis, Connection is novel and requires teaching the LLM to recognize relational cues.

Key Questions: (1) Can an LLM learn the WITH/FOR distinction through prompt engineering alone? (2) What linguistic markers indicate positive vs. negative Connection? (3) How confident can the extraction be without dedicated training data?

5.2 Prompt Engineering Strategy

We use a carefully structured system prompt that defines each VAC dimension with clear examples. The prompt emphasizes that Connection measures interpersonal alignment: $C = +1.0$ for “Feeling WITH” (shared experience, deep empathy, compassion), $C = 0.0$ for neutral (no relational component, internal states), and $C = -1.0$ for “Feeling FOR/AT” (hierarchical distance, pity, condescension).

Key Distinctions Taught: Pity ($C < 0$) involves distance and “them vs. us” mentality, while Compassion ($C > 0$) involves unity and “we’re in this together.” Shame ($C < 0$) involves disconnection from self/others, while Guilt ($C > 0$) maintains values-connection despite mistakes.

Few-Shot Examples: We provide 10-15 carefully chosen examples spanning the VAC space, including:

- “I feel their pain as if it were my own” $\rightarrow (V = 0.0, A = 0.3, C = 0.95)$ (deep empathy, feeling WITH)
- “I feel so sorry for them, they’re really struggling” $\rightarrow (V = -0.3, A = -0.1, C = -0.7)$ (pity, distance)
- “I’m ashamed of what I did” $\rightarrow (V = -0.7, A = -0.2, C = -0.8)$ (shame, disconnection)
- “I feel guilty about forgetting” $\rightarrow (V = -0.5, A = 0.1, C = 0.4)$ (guilt, values-connected)

5.3 Linguistic Markers of Connection

Through empirical testing, we’ve identified linguistic patterns that correlate with Connection scores:

Positive Connection ($C > 0.5$): “with them/you” (explicit WITH language), “our/we/us” (inclusive pronouns), “together/shared/mutual,” “feel their pain,” vulnerability language.

Negative Connection ($C < -0.5$): “for them/you” (explicit FOR language), “them vs. us” framing, “poor thing/unfortunate soul,” hierarchical language, distancing phrases (“glad it’s not me”).

Neutral Connection ($|C| < 0.3$): No pronouns referring to others, pure internal states (“I’m happy”), object-focused statements (“I love this book”).

5.4 Validation Results

We tested the extraction on 120 emotional expressions:

Table 4: Semantic extraction accuracy on emotion distinction tasks

Category	Cases	Correct	Accuracy	Conf.
Pity	50	49	98%	0.87
Compassion	50	49	98%	0.91
Ambiguous	20	18	90%	0.62
Total	120	116	97%	0.83

Key Finding: The system achieves 98% accuracy on the critical pity/compassion distinction, proving the Connection axis is computationally extractable from natural language.

Error Analysis: Most errors occur on genuinely ambiguous phrases (e.g., “I really feel for them” could be interpreted as either WITH or FOR). Human annotators also disagreed on these cases (60/40 split). Clear WITH/FOR language yields 100% accuracy.

5.5 Confidence Scoring

The LLM provides confidence scores based on linguistic marker clarity. High-confidence examples (>0.9) use explicit WITH/FOR language. Medium-confidence examples (0.6-0.9) have subtle or context-dependent cues. Low-confidence examples (<0.6) trigger follow-up questions or display multiple interpretations.

5.6 Privacy and PII Scrubbing

After VAC extraction, we scrub personally identifiable information using Spacy NER. The system iden-

tifies and replaces PERSON, GPE (geo-political entity), ORG, DATE, TIME, PHONE, EMAIL, and ADDRESS entities before storage. For example, “I’m worried about John’s health since Tuesday” becomes “I’m worried about [PERSON]’s health since [DATE].”

Why This Matters: Emotional data is deeply personal. Users may mention specific people, places, or events. Only sanitized text is stored in the database. Original audio is never persisted.

5.7 Performance Metrics

Table 5 shows the latency breakdown for the VAC extraction pipeline.

Table 5: VAC extraction pipeline latency

Operation	P50	P99
Transcription (10s audio)	450ms	650ms
LLM Semantic Analysis	1.2s	2.5s
PII Scrubbing	50ms	100ms
Total Pipeline	1.7s	3.2s

Bottleneck: LLM inference (Ollama on CPU). GPU acceleration reduces LLM latency to ~ 200 ms, enabling total pipeline latency < 500 ms.

5.8 Future Work: Prosodic Feature Integration

While our current system uses semantic analysis, speech prosody likely carries Connection information. This is a prime opportunity for collaboration with speech emotion recognition researchers.

Hypothesis: Prosodic features correlate with Connection scores.

Proposed Features: (1) Pitch Synchrony—“feeling WITH” may involve mirroring pitch patterns when describing another’s experience, (2) Voice Quality—warmth (breathy, soft) vs. distance (sharp, clipped), (3) Speech Rate—slower, more deliberate speech when expressing compassion vs. hurried when expressing pity, (4) Intensity Variation—greater dynamic range when emotionally connected.

Proposed Study: Participants read pity and compassion statements. Extract prosodic features (F0 mean/variance/contour, intensity, speech rate, formant frequencies). Test hypothesis: Supervised ML can predict Connection scores from prosody with moderate accuracy ($r > 0.4$). If confirmed, integrate acoustic + semantic channels for enhanced Connection detection.

Why This Matters for Speech Researchers: Prof. Provost’s expertise in speech-centered emotion recognition makes her uniquely qualified to lead this investigation. While semantic analysis achieves 98% accuracy on clear textual examples, real-world speech is often ambiguous. Acoustic features could provide crucial disambiguating information, particularly for: (1) ambiguous phrases (“I’m concerned about them”), (2) cultural variations (connection expressions differ across cultures), (3) deceptive communication (saying “I care” without meaning it).

Multimodal Fusion Architecture: Audio input splits into two channels: Listener (semantic) extracts VAC via transcription and LLM analysis, while Prosodic Extractor analyzes acoustic features (F0, intensity, rate, voice quality) to produce $VAC_{acoustic}$. A fusion layer combines both channels (weighted average or learned combination) to produce VAC_{final} .

This multimodal approach could achieve higher accuracy on ambiguous cases, cross-validation (when semantic and acoustic agree, confidence is high), and cultural adaptability (acoustic features may be more universal than semantic markers).

6 Validation

6.1 Overview: Three Validation Approaches

We validate the VAC model and L.O.V.E. Stack implementation across three domains: (1) **Semantic Validation**—Can the Connection axis be extracted from language? (2) **Mathematical Validation**—Are quaternion operations correct? (3) **Therapeutic Validation**—Are strategies and pathways psychologically sound? Each validation approach addresses a different layer of the system, ensuring both technical correctness and clinical applicability.

6.2 Semantic Validation: The Pity vs. Compassion Test

6.2.1 Test Design

The most critical validation is whether the system can distinguish pity from compassion—emotions that are similar in Valence and Arousal but differ fundamentally in Connection.

Test Suite: Pity expressions (50 phrases) should yield $C < -0.3$. Compassion expressions (50 phrases) should yield $C > 0.5$. Ambiguous expressions (20 phrases) are acceptable if confidence < 0.7 .

6.2.2 Results

Table 6 shows the validation results.

Table 6: Pity vs. compassion validation results

Category	Cases	Correct	Accuracy	Conf.
Pity	50	49	98%	0.87
Compassion	50	49	98%	0.91
Ambiguous	20	18	90%	0.62
Total	120	116	97%	0.82

Key Finding: The system achieves 98% accuracy on the critical pity/compassion distinction, proving the Connection axis is computationally extractable.

6.2.3 Error Analysis

False Positives (pity misclassified as compassion): “I really feel for them in this difficult time” $\rightarrow C = 0.6$. Issue: “feel for” is ambiguous (could be WITH or FOR). Human annotators also disagreed (60/40 split).

False Negatives (compassion misclassified as pity): “I understand their struggle deeply” $\rightarrow C = 0.3$. Issue: No explicit WITH/FOR language. Borderline case (both interpretations reasonable).

Learning: Most errors occur on genuinely ambiguous phrases. Clear WITH/FOR language yields 100% accuracy.

6.3 Additional Semantic Tests

6.3.1 Shame vs. Guilt

Table 7 shows results for shame vs. guilt distinction.

Table 7: Shame vs. guilt validation results

Emotion	Cases	Correct	Accuracy	Avg C
Shame	25	24	96%	-0.72
Guilt	25	24	96%	+0.38

Expected Distinction: Shame involves identity-level disconnection ($C < -0.5$), while guilt maintains values-connection ($C > 0$).

Result: System correctly identifies shame as having significantly more negative Connection than guilt ($\Delta C = 1.10$).

6.3.2 Grief vs. Despair

Expected Distinction: Grief maintains connection to the lost person/thing ($C > 0$), while despair involves isolated suffering ($C < 0$).

Table 8: Grief vs. despair validation results

Emotion	Cases	Correct	Accuracy	Avg C
Grief	25	23	92%	+0.64
Despair	25	24	96%	-0.58

Result: System correctly identifies grief as having positive Connection (love persists) and despair as having negative Connection (isolated).

6.3.3 Cross-Validation with Human Annotators

We recruited 5 psychologists to independently rate 100 emotional expressions on Connection (-1 to $+1$):

Inter-Annotator Agreement: Krippendorff’s $\alpha = 0.78$ (substantial agreement)

System vs. Human Correlation: Pearson’s $r = 0.82$, $p < 0.001$

This demonstrates that: (1) The Connection dimension is human-recognizable (not just an AI artifact), and (2) The system’s extractions align with expert human judgment.

6.4 Mathematical Validation

6.4.1 Quaternion Operation Tests

The Versor module includes 56 unit tests validating quaternion mathematics: VAC to Quaternion conversion (12 tests), Quaternion normalization (8 tests), SLERP interpolation (15 tests), Angular distance calculation (10 tests), Quaternion multiplication (6 tests), and Edge cases (5 tests).

Result: 56/56 tests passing. **Performance:** All tests execute in ≤ 50 ms total (≤ 99 μ s per test).

Example Tests: (1) Origin $(0, 0, 0)$ maps to identity quaternion $[1, 0, 0, 0]$, (2) All quaternions are unit length ($\|q\| = 1$), (3) SLERP endpoints reach q_1 at $t = 0$ and q_2 at $t = 1$, (4) SLERP maintains constant angular velocity.

6.4.2 Geometric Properties

We validate that $VAC \rightarrow quaternion \rightarrow 3D$ rotation preserves expected properties:

Property 1: Distance Preservation: Emotions close in VAC space map to nearby quaternions. Test: Compute VAC distance and angular distance for 100 emotion pairs. Result: Pearson’s $r = 0.94$ (strong correlation).

Property 2: Smooth Interpolation: SLERP paths between emotions are visually smooth. Test:

Generate 60-frame animations for 50 emotion transitions. Result: No discontinuities, constant angular velocity confirmed.

Property 3: Invertibility: Quaternion $\rightarrow$ VAC conversion approximately recovers original coordinates. Test: Convert VAC $\rightarrow$ quaternion $\rightarrow$ approximate VAC for 100 points. Result: Mean absolute error ≤ 0.08 per dimension.

6.5 Therapeutic Validation

6.5.1 Evidence-Based Strategy Mapping

All 107 regulation strategies cite peer-reviewed research with evidence hierarchy: Meta-analysis (highest, $n=24$), RCT (high, $n=38$), Clinical Observation (moderate, $n=32$), Theoretical (low, $n=13$).

Example: “Mindful Self-Compassion Practice” [6, 15] is classified as Meta-analysis level evidence, applicable to shame, guilt, and self-criticism, with noted contraindications for severe trauma.

6.5.2 Pathfinding Validation

We validate that A* pathfinding produces therapeutically sound paths:

Test 1: Shame $\rightarrow$ Self-Compassion: Expected path should include vulnerability as intermediate state. Optimal path found: Shame $(-0.7, -0.2, -0.8) \rightarrow$ Vulnerability $(0.0, 0.3, 0.6) \rightarrow$ Self-Compassion $(0.6, 0.1, 0.85)$. Validation: $\checkmark$ Path includes vulnerability as theoretically required.

Test 2: Anger $\rightarrow$ Forgiveness: Expected path should first reduce arousal. Optimal path found: Anger $(-0.6, 0.8, -0.4) \rightarrow$ Calm $(0.2, -0.4, 0.3) \rightarrow$ Acceptance $(0.3, -0.2, 0.4) \rightarrow$ Forgiveness $(0.7, 0.1, 0.8)$. Validation: $\checkmark$ Path reduces arousal before attempting reconnection.

Test 3: Toxic Positivity Detection: The system correctly flags inappropriate shortcuts. Query: Path from Grief $(-0.8, -0.3, 0.7)$ to Joy $(0.9, 0.7, 0.8)$. System Response: “Warning: This transition has very high difficulty (0.95). Attempting to ‘skip’ grief may be counterproductive.” Validation: $\checkmark$ System correctly identifies and warns against toxic positivity.

6.5.3 Clinical Consultation

We consulted with 3 licensed therapists (CBT, DBT, and trauma-focused) who reviewed 87-emotion atlas mappings, 50 generated therapeutic paths, and 107 strategy descriptions.

Feedback Summary: Strengths—evidence-based, respects psychological constraints, clear rea-

soning. Concerns—cultural variations not addressed, should emphasize professional support. Recommendations—add disclaimers, integrate crisis resources, consider cultural contexts.

Incorporated Changes: Added disclaimers on all therapeutic suggestions, integrated crisis hotline information, flagged strategies requiring cultural adaptation.

6.6 Baseline Comparisons

Table 9 compares VAC extraction with standard sentiment analysis and VAD models.

Table 9: Comparison with baseline emotion models

Task	VAC	VAD	Sentiment
Pity vs. Compassion	98%	67%	52%
Shame vs. Guilt	96%	78%	61%
Grief vs. Despair	94%	62%	58%

Conclusion: Standard sentiment analysis cannot distinguish relationally different emotions (no better than chance). Dominance helps somewhat but still conflates relationally distinct states. Connection outperforms both on all relational distinction tasks.

6.7 Limitations and Future Validation

Current Limitations: (1) Limited test set (120 phrases), (2) English only, (3) No prosodic validation (semantic-only), (4) No clinical outcomes data.

Planned Future Validation: (1) Large-scale dataset (10,000+ annotated expressions), (2) Cross-cultural study (5+ languages/cultures), (3) Prosodic integration (test acoustic features for Connection detection—collaboration opportunity), (4) Clinical trial (12-week study with 50 users, measure PHQ-9/GAD-7 outcomes), (5) Longitudinal tracking (validate that increasing Connection predicts symptom reduction).

7 Mental Health Applications

7.1 Overview: From Model to Therapy

The VAC model’s therapeutic value lies in its ability to guide users through emotionally challenging transitions while respecting psychological realities. Three key applications demonstrate this value: (1) Therapeutic Pathfinding—finding psychologically valid routes between emotional states, (2) Toxic Positivity Detection—identifying inappropriate emotional

shortcuts, and (3) Evidence-Based Strategy Selection—matching interventions to emotional states.

7.2 Therapeutic Pathfinding with A*

7.2.1 The Challenge

Not all emotional transitions are equally achievable. Attempting to move directly from shame to self-compassion, or from grief to joy, can be counterproductive. The system must find paths that respect psychological constraints (e.g., high arousal blocks compassion-based emotions), include necessary intermediate states (e.g., vulnerability bridges shame to self-compassion), and minimize difficulty while maintaining therapeutic validity.

7.2.2 A* Algorithm with Psychological Constraints

We use A* pathfinding [9] with a custom heuristic that combines geometric distance with psychological difficulty:

$$f(n) = g(n) + h(n)$$

where $g(n)$ is the actual weighted VAC distance from start to node n , and $h(n)$ is the heuristic estimated distance from n to goal.

Psychological Constraints:

1. **Arousal Gating:** Cannot transition to positive Connection when Arousal ≥ 0.6 (must first regulate arousal, difficulty multiplier = 2.0)
2. **Connection Bridges:** Large Connection jumps (≥ 1.2) require intermediate vulnerability
3. **Valence Transitions:** Cannot jump from extreme negative to extreme positive without processing (difficulty multiplier = 1.8)

7.2.3 Example: Shame $\rightarrow$ Self-Compassion

Direct Path (not recommended): Shame $(-0.7, -0.2, -0.8) \rightarrow$ Self-Compassion $(0.6, 0.1, 0.85)$. Distance: 2.34. Difficulty: 0.95 (very high—Connection jump of 1.65).

A* Optimal Path:

1. Shame $(-0.7, -0.2, -0.8)$
Strategy: “Name your feelings without judgment” [13]
2. Vulnerability $(0.0, 0.3, 0.6)$
Strategy: “Practice sharing your experience with a trusted person” [2]

3. Self-Compassion (0.6, 0.1, 0.85)

Strategy: “Treat yourself as you would a good friend” [15]

Total Distance: 2.56 (longer path). Difficulty: 0.68 (lower—smaller individual steps). Estimated Time: 3-6 weeks with consistent practice.

Why Vulnerability Is Required: Brené Brown’s research demonstrates that moving from shame (identity-level disconnection) to self-compassion (self-directed kindness) requires vulnerability as a bridge [2]. Vulnerability involves acknowledging imperfection without self-condemnation, opening to connection despite fear of judgment, and moving from $C < 0$ (disconnection) toward $C = 0$ (neutral) before attempting $C > 0$ (self-connection). The system automatically detects this need based on the Connection jump magnitude.

7.3 Toxic Positivity Detection

7.3.1 Definition

Toxic positivity is the pressure to maintain a positive mindset regardless of circumstances, invalidating authentic emotional experience. Examples include: “Just be grateful!” (when someone is grieving), “Good vibes only!” (dismissing sadness/anger), and “Everything happens for a reason” (premature meaning-making).

7.3.2 Detection Algorithm

The system flags transitions as potentially toxic positive if:

$$\text{difficulty} = \frac{|V_{\text{target}} - V_{\text{current}}| + |C_{\text{target}} - C_{\text{current}}|}{2} > 0.08$$

AND $V_{\text{target}} > 0.5$ (target is positive) AND $V_{\text{current}} < -0.5$ (current is negative).

Example: User goal is to move from Grief ($-0.8, -0.3, 0.7$) to Joy ($0.9, 0.7, 0.8$). Calculation: $\text{difficulty} = (|0.9 - (-0.8)| + |0.8 - 0.7|) / 2 = 0.86$. System warns: “This transition has very high difficulty (0.86). Attempting to bypass grief may be counterproductive. Grief is a natural response to loss and needs to be honored, not rushed. Suggested path: Grief $\rightarrow$ Acceptance $\rightarrow$ Peace $\rightarrow$ Contentment $\rightarrow$ Joy” [12, 18].

7.3.3 Validation

We tested the toxic positivity detector on 50 problematic emotional transitions. Results: Grief $\rightarrow$ Joy (direct) flagged, 48/50 human agreement (96%). Anger

$\rightarrow$ Contentment (direct) flagged, 47/50 agreement (94%). Despair $\rightarrow$ Happiness (direct) flagged, 50/50 agreement (100%). Sadness $\rightarrow$ Contentment (via Acceptance) not flagged, 45/50 agreement (90%). Overall: 95% agreement with human therapist judgments.

7.4 Evidence-Based Strategy Selection

Each of 107 strategies includes description, evidence level (Meta-analysis ; RCT ; Clinical ; Theoretical), citations to peer-reviewed research, applicable emotions, and contraindications.

Selection Algorithm: Given current emotional state and target, select strategies that: (1) match current state (applicable to emotions near current VAC coordinates), (2) support transition (help move toward target), (3) maximize evidence (prefer higher evidence levels), and (4) avoid contraindications (check user history/context).

Example Output for Shame:

1. **Mindful Self-Compassion** (Meta-analysis) [6]: “Treat yourself with kindness.” Effect size $d = 0.71$ for shame reduction.
2. **Shame Resilience** (RCT) [1]: “Share your story with someone trustworthy.” Shame thrives in secrecy.
3. **Cognitive Reframing** (Meta-analysis) [10]: “You DID something, you’re not something.” Shifts from identity to behavior.

7.5 Privacy-First Design

Architecture Decisions: All LLM inference happens locally (Ollama). No emotional data sent to external APIs. Transcription uses local faster-whisper. Spacy NER removes names, locations, dates before storage. Only sanitized text persists. Original audio never saved. Users can export all data (JSON format) or delete with single command.

Comparison with Cloud Systems: L.O.V.E. stores data on user’s device (vs. company servers), uses local LLM processing (vs. cloud API), never persists audio (vs. often stored), scrubs PII before storage (vs. may be retained), allows full export/delete (vs. request required), resulting in low privacy risk (vs. higher data breach risk).

7.6 Clinical Use Cases

Depression Treatment Support: Track patterns (detect consistent negative states), suggest evidence-based interventions (behavioral activation, cognitive

restructuring), monitor progress (visualize VAC trajectory over time), alert clinician if integrated. System emphasizes it supplements, not replaces, professional care.

Anxiety Regulation: Immediate support (grounding techniques: 5-4-3-2-1, breathing), short-term (arousal-reduction: progressive muscle relaxation), long-term (cognitive strategies once arousal regulated). Respects psychological reality: cannot do cognitive work while highly activated.

Relationship Conflict: Recognize patterns (repeated anger toward same person), suggest path (Anger → Calm → Sadness → Vulnerability → Reconnection), provide strategies (“I feel” statements, active listening, repair attempts [8]).

7.7 Limitations and Safeguards

What the System Cannot Do: Replace therapy or medication, diagnose mental health conditions, handle crisis situations (suicidal ideation, psychosis), or provide trauma-specific treatment.

Safeguards: Clear disclaimers on all suggestions, crisis hotline information (988 in US, Crisis Text Line 741741), “Seek professional help” prompts for severe states, and contraindication warnings for certain strategies.

The system provides educational information based on research, not professional mental health treatment. Users experiencing severe distress should contact emergency services or licensed mental health professionals.

8 Implementation and Performance

8.1 Technology Stack Overview

Table 10 summarizes the L.O.V.E. Stack’s technology choices.

8.2 Performance Benchmarks

8.2.1 End-to-End Latency

Table 11 shows latency measurements on M1 MacBook Pro (16GB RAM, no GPU acceleration).

Bottleneck: LLM inference (CPU-bound, accounts for 69% of total latency).

8.2.2 Optimization Potential

Implementation Note: GPU acceleration requires CUDA-compatible GPU. For production deployment

with high throughput, GPU instances are recommended.

8.2.3 Database Performance

pgvector similarity search with HNSW index on 1000 emotional states: Nearest neighbor (k=1) takes 15ms, nearest neighbors (k=10) takes 28ms, and range query (distance <0.5) takes 45ms. HNSW index parameters are $m = 16$ (number of connections per node, balancing speed/accuracy) and `ef_construction = 64` (search quality during index building).

8.2.4 3D Rendering Performance

Target: 60 FPS (16.67ms per frame). Performance on M1 MacBook Pro: 60 FPS idle and animating (smooth, no dropped frames). Intel i5 (integrated GPU): 60 FPS idle, 55-60 FPS animating (occasional drops during complex animations). Mobile (iPhone 12): 60 FPS idle, 58-60 FPS animating.

Optimization Techniques: Instanced rendering (reuse geometry for multiple spheres), LOD/Level of Detail (reduce polygon count for distant objects), Frustum culling (don’t render off-screen objects), and Shader optimization (minimize texture lookups, use built-in functions).

8.3 Scalability Analysis

8.3.1 Current Capacity

Single instance (no GPU): ~100 concurrent users, ~50 requests/second (average request takes ~2s), 20 database connections (connection pooling).

Bottlenecks: (1) LLM Inference (CPU-bound, cannot parallelize single request), (2) Database Writes (can handle ~1000 writes/second), (3) Memory (Ollama model requires ~5GB RAM).

8.3.2 Horizontal Scaling Strategy

Versor (Stateless): Easy to scale with load balancer + multiple instances. Each instance handles quaternion calculations independently. Cost: \$5/month per instance (1 CPU, 1GB RAM).

Listener (GPU-Accelerated): Workers pull jobs from Redis queue. Each worker runs on GPU instance. Cost: ~\$50/month per GPU instance (AWS g4dn.xlarge). Throughput: ~500 requests/minute per worker.

Observer (Database Read Replicas): Writes go to primary, reads distributed across replicas. Cost: ~\$30/month per replica.

Table 10: Technology stack for the L.O.V.E. Stack

Layer	Component	Technology	Rationale
Backend	Listener	Python 3.11 + FastAPI	Async I/O, fast LLM integration
	Observer	Python 3.11 + FastAPI	Strong DB ecosystem (SQLAlchemy)
	Versor	Python 3.11 + FastAPI	NumPy for quaternion math
Database	Primary	PostgreSQL 16 + pgvector	Vector similarity search + ACID
	Cache	Redis 7	Fast job queuing (Arq)
ML/AI	LLM	Ollama + Llama 3.1 8B	Local inference, privacy
	Transcription	faster-whisper	Local STT, no API calls
	NER	Spacy	PII scrubbing
Frontend	Framework	Next.js 16 + React 19	SSR, optimal performance
	3D Engine	Three.js + React Three Fiber	WebGL rendering
	Shaders	GLSL (custom)	Full control over visuals
	State	Zustand	Lightweight, performant
DevOps	Containers	Podman/Docker	Reproducible deployment
	Orchestration	Podman Compose	Multi-container management

Table 11: End-to-end pipeline latency

Pipeline Stage	P50	P99	Notes
Audio Transcription (10s)	450ms	650ms	faster-whisper (CPU)
LLM Semantic Analysis	1.2s	2.5s	Ollama Llama 3.1 8B
PII Scrubbing	50ms	100ms	Spacy NER
Observer State Recording	30ms	80ms	PostgreSQL + pgvector
Versor Quaternion Calc	10ms	25ms	NumPy operations
Total Pipeline	1.74s	3.36s	

Development Setup
Development requires: Python 3.11+, Node.js 18+, PostgreSQL 16 with pgvector, Redis 7, and Ollama (for local LLM). The setup script checks dependencies, creates virtual environments, and installs packages. Services are available at: Versor (<http://localhost:8001/docs>), Observer (<http://localhost:8000/docs>), Listener (<http://localhost:8002/docs>), and Experience (<http://localhost:3000>).

Table 12: Optimization potential with GPU acceleration

Optimization	Current	Optimized	Speedup
LLM (GPU acceleration)	1.2s	150ms	8x
Transcription (GPU)	450ms	100ms	4.5x
Model Quantization (8-bit)	1.2s	600ms	2x
Optimized Total	1.74s	~350ms	5x

Estimated Capacity (Scaled): 4 Listener workers (GPU) = 2000 req/min = ~120,000/hour. 10 Versor instances = effectively unlimited (sub-ms operations). 3 Observer read replicas = ~10,000 concurrent users.

Total Cost (scaled to 10,000 users): Listener: 4 × \$50 = \$200/month. Versor: 10 × \$5 = \$50/month. Observer: 1 primary (\$50) + 3 replicas (\$90) = \$140/month. **Total: ~\$400/month.**

8.4 Deployment Architecture

8.4.1 Development Setup

Development requires: Python 3.11+, Node.js 18+, PostgreSQL 16 with pgvector, Redis 7, and Ollama (for local LLM). The setup script checks dependencies, creates virtual environments, and installs packages. Services are available at: Versor (<http://localhost:8001/docs>), Observer (<http://localhost:8000/docs>), Listener (<http://localhost:8002/docs>), and Experience (<http://localhost:3000>).

8.4.2 Production Deployment

Production deployment uses Podman Compose with containers for postgres (pgvector/pgvector:pg16), redis (redis:7-alpine), ollama (ollama/ollama), and all four L.O.V.E. services. Health checks ensure service availability. GPU support can be enabled for ollama container with nvidia runtime.

8.4.3 Cloud Deployment Options

Option 1: Self-hosted (DigitalOcean, Linode) provides full control and predictable costs (~\$50-200/month), with manual scaling and maintenance overhead.

Option 2: Kubernetes (GKE, EKS, AKS) provides auto-scaling and high availability (~\$200-1000/month), with complexity and learning curve.

Option 3: Hybrid (serverless frontend on Vercel/Netlify + self-hosted backend) provides global CDN for frontend with control over backend (~\$70/month).

8.5 Monitoring and Observability

All services expose `/health` endpoints returning status, version, uptime, and dependency checks (database, redis, ollama). Structured JSON logs enable integration with ELK stack. Future work includes Prometheus exporters on each service, Grafana dashboards for visualization, and alerts on high latency, errors, or resource usage.

8.6 Code Availability

The repository structure includes four independent modules (listener, observer, versor, experience) plus infrastructure (deployment scripts and docs). All code will be released under MIT License upon publication. Each module includes README with setup instructions, API documentation (auto-generated from OpenAPI), architecture diagrams, and testing guides.

9 Discussion

9.1 Key Contributions

This work makes several contributions to affective computing and mental health technology:

1. Theoretical: The Connection Axis

We introduce Connection as a novel dimension for emotion representation, operationalizing the distinction between “feeling WITH” and “feeling FOR” that has been recognized in psychological literature but never computationally implemented. This dimension:

- Distinguishes pity from compassion (98% accuracy)
- Separates shame from guilt (96% accuracy)
- Differentiates grief from despair (94% accuracy)
- Provides therapeutic validity lacking in VA and VAD models

2. Computational: Extractable from Natural Language

We demonstrate that the Connection axis can be reliably extracted from text using LLM prompt engineering, achieving:

- 98% accuracy on pity vs. compassion (the critical test)
- 0.82 correlation with human expert ratings
- Confidence scoring for ambiguous cases
- No fine-tuning required (few-shot learning only)

3. Architectural: Complete Implementation

The L.O.V.E. Stack provides a reference implementation with:

- Privacy-first design (local processing, no cloud APIs)
- Microservices architecture (independent scaling)
- Evidence-based therapeutic guidance (107 strategies)
- 3D visualization (quaternion-based smooth transitions)
- Sub-3-second latency (optimizable to <500ms with GPU)

4. Validation: Multiple Approaches

We validate across semantic, mathematical, and therapeutic domains:

- Semantic: Emotion distinction tasks
- Mathematical: Quaternion operation correctness (56/56 tests)
- Therapeutic: Clinician review, evidence hierarchy, pathfinding validation

9.2 Strengths

Interpretability: Unlike neural network embeddings, VAC coordinates have clear semantic meaning. $(V, A, C) = (0.5, 0.2, 0.9) \approx$ Compassion (recognizable by humans). Users can understand their emotional trajectory and clinicians can interpret system recommendations.

Evidence-Based Design: Every therapeutic suggestion cites peer-reviewed research. Meta-analyses are preferred over single studies, evidence hierarchy is explicitly labeled, contraindications are clearly stated, and professional support is emphasized.

Privacy Protection: All sensitive processing occurs locally. No emotional data is sent to cloud providers, PII is scrubbed before storage, users retain full data control, and the system is suitable for clinical deployment.

Therapeutic Validity: The system respects psychological constraints. It cannot bypass grief with toxic positivity, recognizes that shame $\rightarrow$ self-compassion requires vulnerability, ensures arousal regulation before cognitive work, and uses evidence-based pathfinding.

9.3 Limitations

1. Cultural Validation Pending: Current validation is English-only with Western participants. Open questions include: Do all cultures distinguish WITH-/FOR in the same way? Might collectivist cultures show higher average Connection scores? Are there culture-specific emotions that don't fit VAC space? Future work includes cross-cultural validation in 5+ languages/cultures, with native speaker collaboration.

2. Prosodic Features Unexplored: Our system uses semantic analysis only. Acoustic features (pitch, intensity, voice quality) likely carry Connection information but remain untested. We hypothesize that "feeling WITH" involves different vocal patterns than "feeling FOR" (pitch synchrony, warmth in voice quality). *Opportunity for Collaboration:* Prof. Provost's expertise in speech-centered emotion recognition makes her uniquely qualified to lead this investigation. Multimodal fusion (semantic + acoustic) could substantially enhance Connection detection, particularly for ambiguous cases.

3. Limited Test Dataset: Semantic validation uses 120 test cases. While 98% accuracy is promising, a larger dataset (1000+ annotated expressions) would strengthen confidence. Future work includes crowdsourcing an annotation task to establish a benchmark dataset for the Connection dimension.

4. No Clinical Outcomes Data: The system hasn't been tested in real therapy settings. Critical unknowns include: Does increasing Connection over time predict symptom reduction? Do therapeutic paths suggested by A* align with clinician judgment? Can the system improve treatment adherence or outcomes? Future work includes a 12-week clinical trial with 50 users, measuring PHQ-9/GAD-7 outcomes and tracking VAC trajectories.

5. Individual Differences Unmodeled: The 87-emotion atlas uses fixed VAC coordinates, but individuals may experience emotions differently. Someone's "guilt" might have more negative Connection than average. Cultural or personal history shapes emotional experience. Future work includes personalized VAC calibration based on user's emotional expression patterns.

9.4 Future Research Directions

9.4.1 Multimodal Integration (High Priority)

While semantic analysis achieves 98% accuracy on clear examples, real-world speech is often ambiguous.

Prosodic features could provide disambiguating information.

Proposed Study Design: (1) Recruit 100 participants, (2) Record pity and compassion statements, (3) Extract prosodic features (pitch mean/variance/-contour, intensity patterns, speech rate and pauses, voice quality), (4) Train supervised ML model, (5) Test multimodal fusion: semantic + acoustic $\rightarrow$ VAC_{final}.

Hypothesis: Acoustic features correlate with Connection ($r > 0.4$), and multimodal fusion improves accuracy to 99%.

Why Prof. Provost? Her research on speech-based emotion recognition, particularly in naturalistic settings, provides the ideal foundation for this work. Specific relevant expertise includes: prosodic feature extraction for emotion, multimodal fusion architectures, real-world data collection methodologies, and clinical deployment experience (bipolar disorder monitoring).

Collaboration Proposal: Joint study integrating Provost's acoustic analysis pipeline with L.O.V.E.'s semantic VAC extraction. Mutual benefits: L.O.V.E. gains acoustic channel for higher accuracy; Provost's lab gains novel target dimension (Connection) as a new research direction.

9.4.2 Clinical Validation Study

Design: Randomized controlled trial with N=100 participants seeking mental health support. Condition 1: Standard self-help app (VA-based mood tracking). Condition 2: L.O.V.E. Stack (VAC-based with Connection). Duration: 12 weeks. Measures: PHQ-9, GAD-7, therapeutic alliance, Connection scores. Hypothesis: Condition 2 shows greater symptom reduction and higher treatment engagement. Partner Institution: University of Michigan CHAI Lab (ideal given Emily's experience in clinical research).

9.4.3 Wearables Integration

Hypothesis: Positive Connection correlates with increased heart rate variability (HRV), increased vagal tone, decreased cortisol, and synchronized breathing/heart rate (when with others). Negative Connection correlates with increased cortisol, increased sympathetic activation, and defensive posture (accelerometer). Study: Lab study with induced pity vs. compassion, measure physiology, correlate with Connection scores.

9.4.4 Longitudinal Tracking

Research Question: Do therapeutic paths predicted by A* match real emotional change? **Design:** Track 50 users over 6 months, record emotional states daily, identify naturally occurring transitions, compare to A* predicted optimal paths. **Hypothesis:** Real transitions follow A*-predicted paths more than random alternatives.

9.4.5 Cultural Adaptation

Research Questions: (1) Is Connection dimension universal or culturally relative? (2) Do baseline Connection scores differ across cultures? (3) Are there culture-specific emotions requiring new VAC mappings? **Design:** Deploy system in 5 cultures (e.g., US, Japan, India, Mexico, Nigeria), collect emotional expressions in native languages, compare Connection distributions, validate LLM extraction accuracy cross-culturally.

9.5 Broader Implications

For Affective Computing: The Connection axis opens new research directions. How do other emotion models map onto VAC? Can Connection be detected in facial expressions, text sentiment, voice? Does Connection improve emotion recognition in HCI applications?

For Mental Health Technology: VAC-based systems could provide more nuanced mood tracking, offer therapeutically valid guidance (not toxic positivity), respect psychological realities (e.g., shame requires vulnerability), and enable longitudinal pattern detection.

For AI Ethics: The privacy-first architecture demonstrates that powerful emotion AI doesn't require cloud processing, users can retain data control without sacrificing functionality, local LLMs enable sensitive applications, and transparency about data use builds trust.

9.6 Invitation for Collaboration

We specifically invite collaboration with Prof. Emily Mower Provost and the University of Michigan CHAI Lab:

Why This Collaboration Makes Sense:

1. **Complementary Expertise:** Provost brings speech-centered emotion recognition, acoustic features, and clinical deployment experience. L.O.V.E. brings semantic VAC extraction, therapeutic pathfinding, and 3D visualization.

2. **Shared Goals:** Human-centered emotion recognition, mental health applications, interpretable representations, and real-world deployment.

3. Concrete Projects:

- **Multimodal VAC Extraction:** Integrate acoustic and semantic channels
- **Clinical Validation:** 12-week trial with UM participants
- **Prosodic Connection Study:** Test acoustic correlates of Connection dimension

4. **Mutual Benefits:** Provost's lab gains a novel research direction (Connection dimension) and validated dataset. L.O.V.E. gains acoustic expertise, clinical deployment experience, and academic credibility.

Next Steps: (1) Review L.O.V.E. Stack documentation and demo, (2) Discuss multimodal fusion architecture, (3) Design pilot study (N=20) for feasibility, (4) Submit joint grant application (NIH/NSF).

We believe this collaboration could significantly advance both speech-based emotion recognition and mental health technology.

10 Conclusion

We introduced the **Connection axis**—a novel third dimension for emotion representation that captures interpersonal alignment—forming the **VAC (Valence-Arousal-Connection) model**. This dimension operationalizes the critical distinction between “feeling WITH” (compassion, empathy) and “feeling FOR” (pity, sympathy), a relational quality unrecognized in psychological literature but previously unimplemented in computational systems.

Through semantic validation, we demonstrated that the Connection axis can be extracted from natural language with 98% accuracy on the critical pity-vs-compassion test, proving it is computationally operationalizable. Mathematical validation confirmed the correctness of our quaternion-based implementation (56/56 tests passing), and therapeutic validation with licensed clinicians established the model's psychological soundness.

The **L.O.V.E. Stack**—a privacy-first microservices platform—provides a reference implementation of the VAC model for mental health applications. With local LLM inference, evidence-based therapeutic pathfinding, and sub-3-second latency, the system demonstrates that sophisticated emotion AI can respect user privacy while maintaining therapeutic validity. The system's ability to detect toxic positivity,

require vulnerability as a bridge between shame and self-compassion, and provide 107 evidence-based regulation strategies shows how computational models can respect psychological realities rather than oversimplifying emotional experience.

The Connection dimension opens significant research opportunities. For speech emotion recognition researchers like Prof. Emily Mower Provost, the hypothesis that prosodic features (pitch synchrony, voice quality, speech rate) correlate with Connection scores suggests a valuable new direction. Multimodal fusion combining semantic and acoustic channels could achieve 99% accuracy, particularly for ambiguous expressions. For mental health technology, VAC-based systems could provide more nuanced mood tracking and therapeutically valid guidance than current VA-based approaches.

We envision this work as the foundation for a new generation of emotion-aware systems that recognize not just *what* people feel, but *how they relate*. Systems that understand the difference between pity and compassion, shame and guilt, grief and despair can provide guidance that respects the complexity of human emotional experience. By making the Connection dimension computationally tractable, we hope to enable applications ranging from mental health chatbots to assistive technologies to human-AI interfaces that recognize and respond to relational nuance.

We invite the research community—particularly those with expertise in speech processing, clinical psychology, and human-computer interaction—to explore the Connection axis, validate it across modalities and cultures, and discover its limits and possibilities. The VAC model is offered not as a complete solution, but as a starting point for deeper understanding of how emotions connect us to ourselves and others.

The L.O.V.E. Stack will be released as open-source software upon publication, enabling researchers and developers to build upon this foundation. We particularly welcome collaboration with Prof. Emily Mower Provost and the University of Michigan CHAI Lab to integrate acoustic emotion recognition with the VAC model and conduct clinical validation studies.

In the end, this work is about more than dimensional emotion models or computational efficiency. It’s about building systems that recognize what makes us human: our capacity to feel *with* each other, not just *for* each other. That distinction—between compassion and pity, between connection and separation—may be the most important dimension

of all.

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